Logical Operators

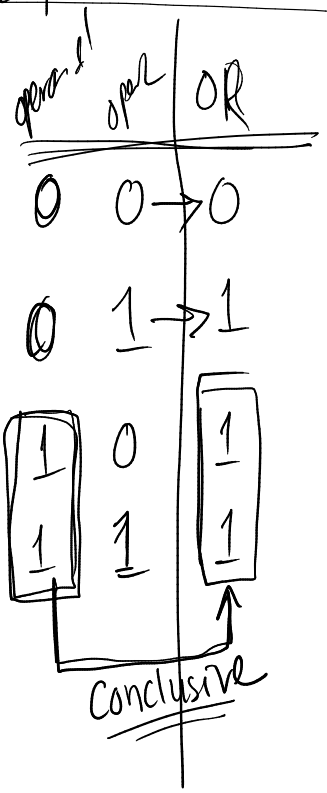
&& AND || OR ! NOT

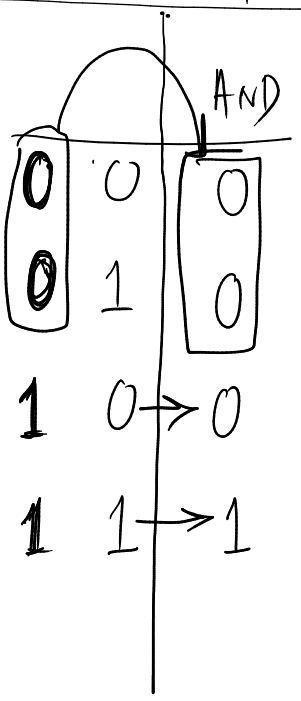
⟵ Binary ⟶ Unary operator

```
int a = 3, b = 0, c = -5;

Pf("%d", a && b) // 0
   , b || c) // 1
   , !b && c) // 1
```

		AND	OR
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1





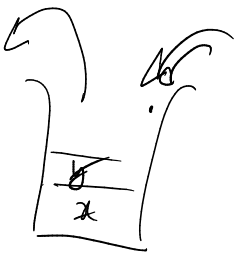
Reverse Polish Notation (RPN): Infix → Postfix → Evaluate

↳ To have a Single solution.

x + y - z

stack → [x, y, +, z, -]

stack | Evaluate



347 + 678 - 378

x + y + z

Full - Grouping Using Precedence and associativity

- Execution: left → Right

X || Y && Z ⇒ (X || (Y && Z))

Skip

b = a*b/(a=b); ⇒ b=((a*b)/(a=b))

int a, b, c;

≡

result = (a==b) && (b==c);

→ 1: All the three variables contain same value

→ 0;

for some value of a, b, and c

result = (a<0) || (b<0) || (c<0);
F F F

→ 0 : a, b, c all are +ve Data

→ 1 : at least one of the three variable is -ve

≡

r = (a>b) && (b>c);

→ 1 : a is the higher among the three, and c is the lowest

→ 0 ;

---

char ch;

≡

r = (ch >= 97) && (ch <= 122);

→ 1 : ch is lowercase alphabet

→ 0 ;

97 — ch — 122
'a' 'z'

---

r = (ch >= 'A') && (ch <= 'Z')

→ 1 : ch contains uppercase alphabet

ch = getchar( ); // scanf("%c", &ch);

// how will you check, ch is Alphabet or not ?

65 — 90 , 91 — 96 , 97 — 122

r = (upp) || (low);

→ 1: Alphabet — ((ch >= 'A') && (ch <= 'Z')) || ((ch >= 'a') && (ch <= 'z'))

→ 0: Not Alphabet

Conditional operator (? :)

— the only ternary operator.

→ (test condition) ? on true : on false ;

c = ((a == b) ? ++a : ++b);

C = ++a else C = ++b

( ) ? : ;

variable, constant, functioncall, expression

Can't use keywords

W.A.P to check the Supplied character is alphabet or not

I/P	O/P
'A'	Yes Alp
'5'	NO
*	NO,
b	Yes

x && y → logical AND

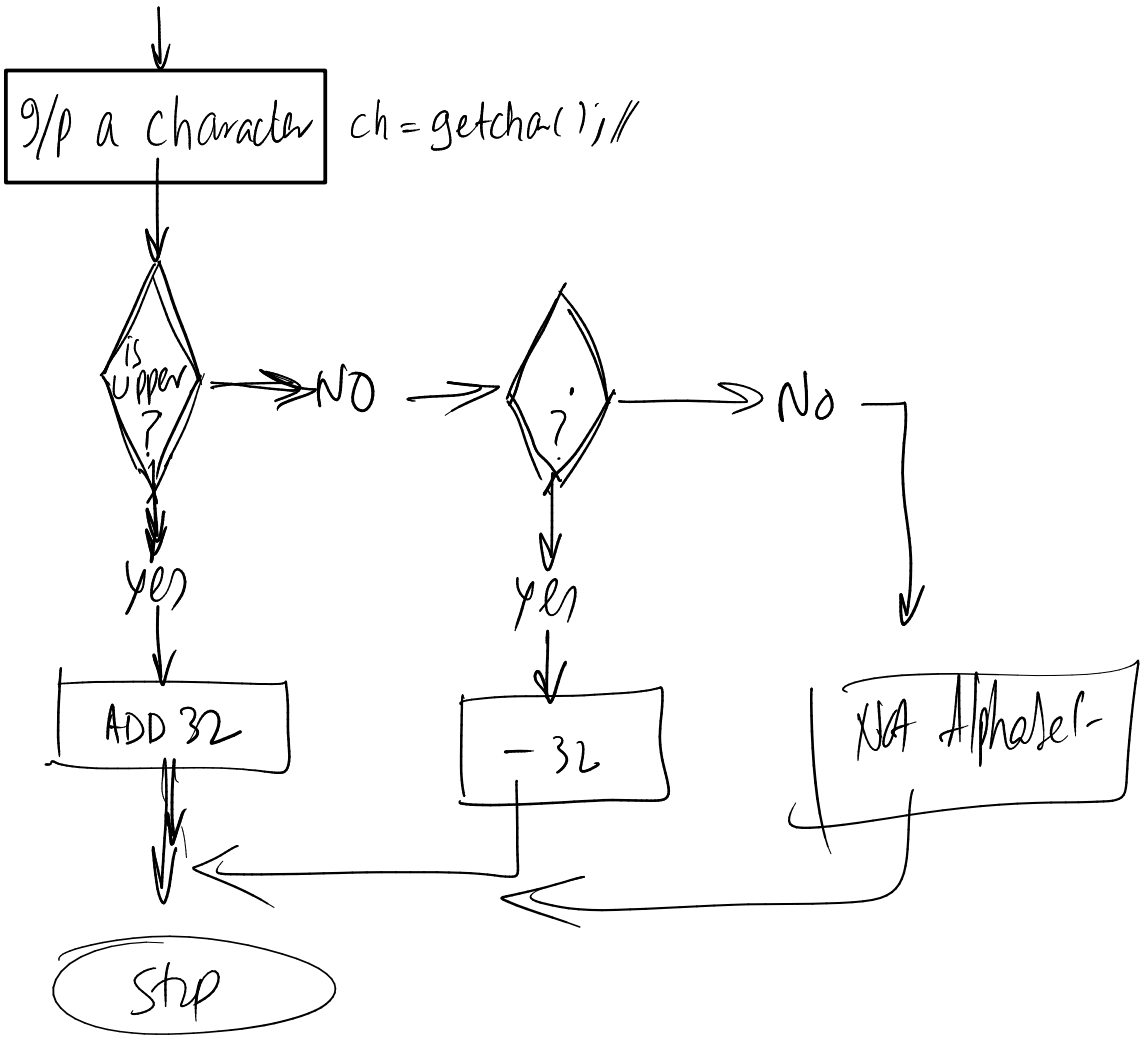
x & y → bitwise AND

&y → reference operator or Address of operator

scanf("%c", &v);

will wait for Input, from stdIn Buffer.

If data available, then collects and returns.



Nesting

(c1) ? : ;

(c2) ? : ;

(C1) ? _ : (c2) ? ( ) : ( );

((ch >= 65) && ch <= 90) ? ch += 32 : ((ch >= 97) && (ch <= 122) ? ch -= 32 : Pf("Not ");

⇓

ch = ch + 32;

ch = getchar();

Task to complete by tomorrow

dummy

(ch >= 48) && (ch <= 57) ? pf("Numeric char"); 0;

(ch >= 65) && (ch <= 90) ? pf("Uppercase Alphabet"); 0;

(ch >= 97) && (ch <= 122) ? pf("lower case alphabet"); 0;

(... ? ...) ? pf("spl char"); 0;

2nd task

Use Single (Nested) ~~conditi~~ statement.

